ABSTRACT

Particularly GPT-4 and LLaMA, recent developments in foundation models (FMs) have drawn a lot of attention. In zero-shot learning contexts, these models have displayed excellent performance. Models like Grounding DINO and the Segment Anything Model (SAM) have excelled in open-set detection and instance segmentation tasks in the field of visual learning. Their accomplishments suggest that these FMs will significantly influence practical visual learning tasks, bringing about a paradigm change in model construction. To demonstrate its efficacy, we share preliminary findings from our datasets. With the least amount of model tuning, our approach aims to maximise the applicability of visual FMs in certain scenarios.

The remote sensing field, which deals with images different from typical circumstances, is the subject of this study. We have created a pipeline called Text2Seg that makes use of a number of FMs to simplify tasks involving the semantic segmentation of remote sensing images that are prompted by text. Benchmarking our process to popular remote sensing

Keywords: Semantic segmentation, Visual Foundation Model, and Remote Sensing

INTRODUCTION

Large-scale foundational language models have seen a boom in research in recent years, and these models exhibit amazing zero-shot generalisation abilities in a variety of downstream applications. These models have been successful in a variety of industries, including public health, education, estimating global warming, and healthcare. Researchers have started looking into the theoretical underpinnings of visual learning in response to this achievement. The initial focus of visual foundation models was on producing linguistically or numerically sophisticated representations of visuals. The foundation models for fine-grained tasks like semantic segmentation, which call for a deeper comprehension of the geometric structure of images, have, nevertheless, been the focus of more recent study.

Semantic segmentation is essential in the field of remote sensing for tasks like estimating land use and monitoring the environment. It entails Accurately separating geographical items (such as grass, trees, buildings, water bodies, and cars) from background in airborne sensor photos, UAV (Unmanned Aerial Vehicle) photographs, or satellite images. To handle inconsistencies caused by several sensors, conventional methods entailed training distinct models on various datasets. These models' robustness and generalizability, however, are frequently compromised when used to test data gathered under various settings. Consequently, a semantic segmentation model that consistently outperforms competing approaches is required. The Segment Anything Model (SAM), trained on a sizable dataset with billions of masks, was presented as the first fundamental model for object segmentation.